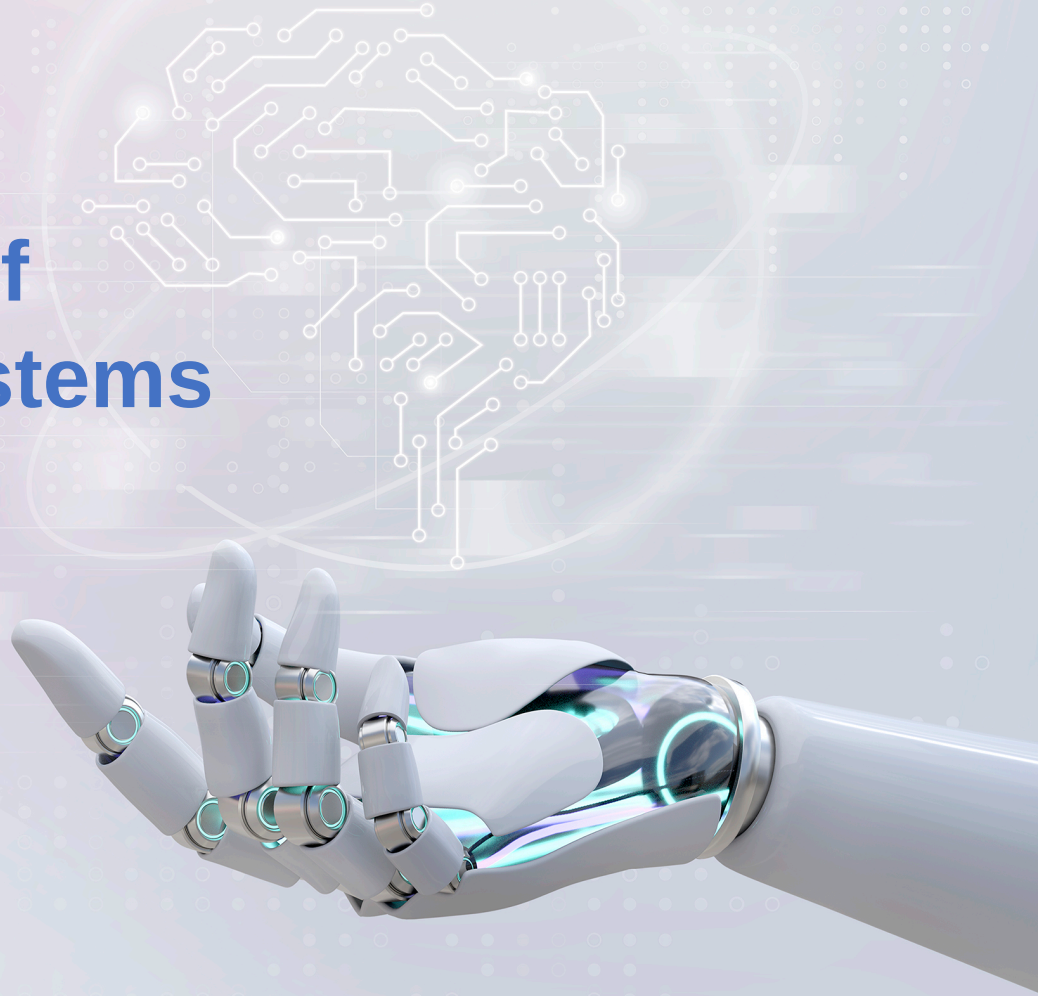
AI Compute Faculty Development Initiative

[Overview] Hardware & Software Fundamentals of Intelligent Computing Systems

Guojie Luo
罗国杰
gluo@pku.edu.cn

Jing Wang
王晶
jwang@ruc.edu.cn

October 2026 · 2nd AICFDI



Supported by UDNP.

Outline

1. **Opening** — motivation, CAAFDP, FlagOS
2. **Modules** — five-module map, Oct 12–25 schedule, previews
3. **Practice & close** — labs, takeaways, resources

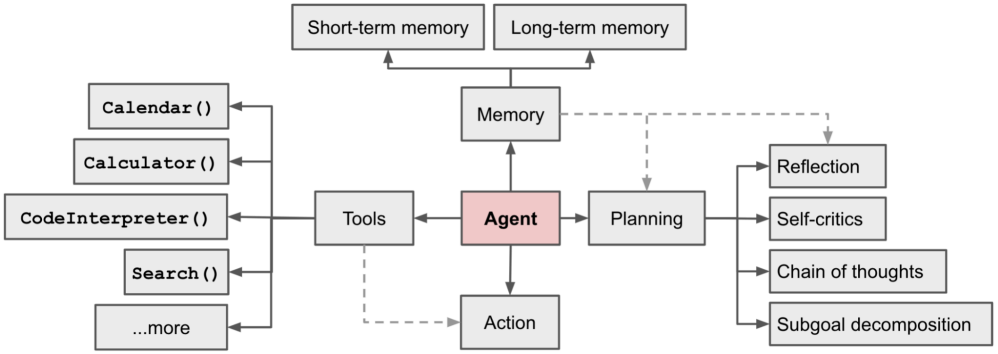
1. Opening

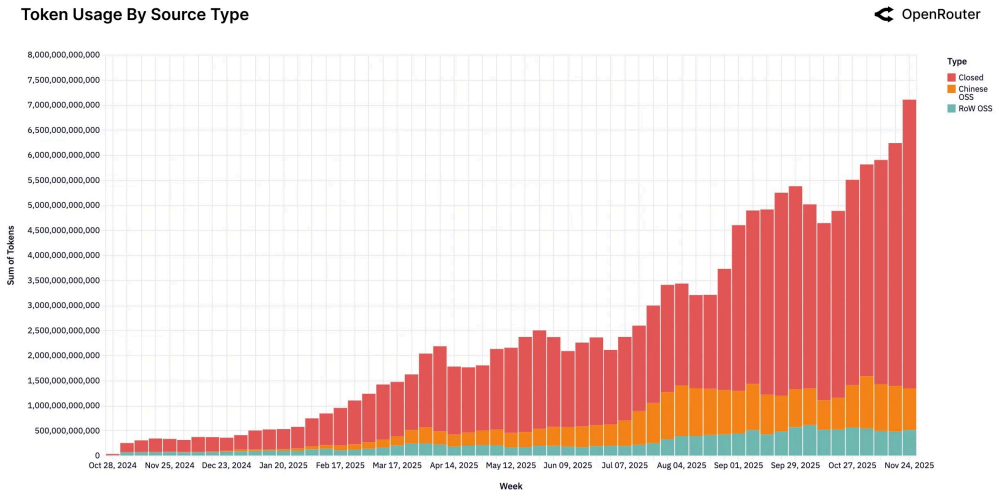
- Why intelligent computing systems?
- AICFDI context & FlagOS
- Five-module map & October schedule

1.1. Token-Hungry Generative AI

- Agents, coding copilots, multimodal generation — **tokens as the new workload unit**
- Demand grows faster than single-device capacity
- Systems must deliver **effective** FLOPS and **effective** bandwidth together

Not another GPU/NPU programming course — present LLM systems the way how “computer organization” course does.





1.2. What Counts as “General-Purpose”?

- By #programs → CPU is (still) general-purpose
- By TOP/s → GPU / NPU are general-purpose for AI

Specialization splits computing into a fast lane and a slow lane

- Technology that advances attracts users and capital;
- stalled specialization stalls ecosystems.

B. N. C. Thompson & S. Spanuth, “The Decline of Computers as a General Purpose Technology,” MIT IDE Research Brief, 2019.

1.3. Model Scale vs Device Memory

NLP “Moore’s Law”

Parameters: $> 10,000 \times$ in 7 years

Year	Model	Size
2018	BERT-Large	0.34 B
2020	GPT-3	175 B
2024	Llama-3.1	405 B
2024	DeepSeek-V3	671 B MoE
2025	GPT-4.5	≈12.8 T MoE

GPU memory growth

Capacity: $< 10 \times$ in 7 years

Year	GPU	Mem
2018	V100	32 GB
2020	A100	80 GB
2023	H100	94 GB
2024	H200	141 GB
2025	B200	192 GB

Multi-device / multi-node parallelism and high-bandwidth interconnects are **mandatory**, not optional.

1.4. 1st AI Compute Faculty Development Initiative (April 2026)







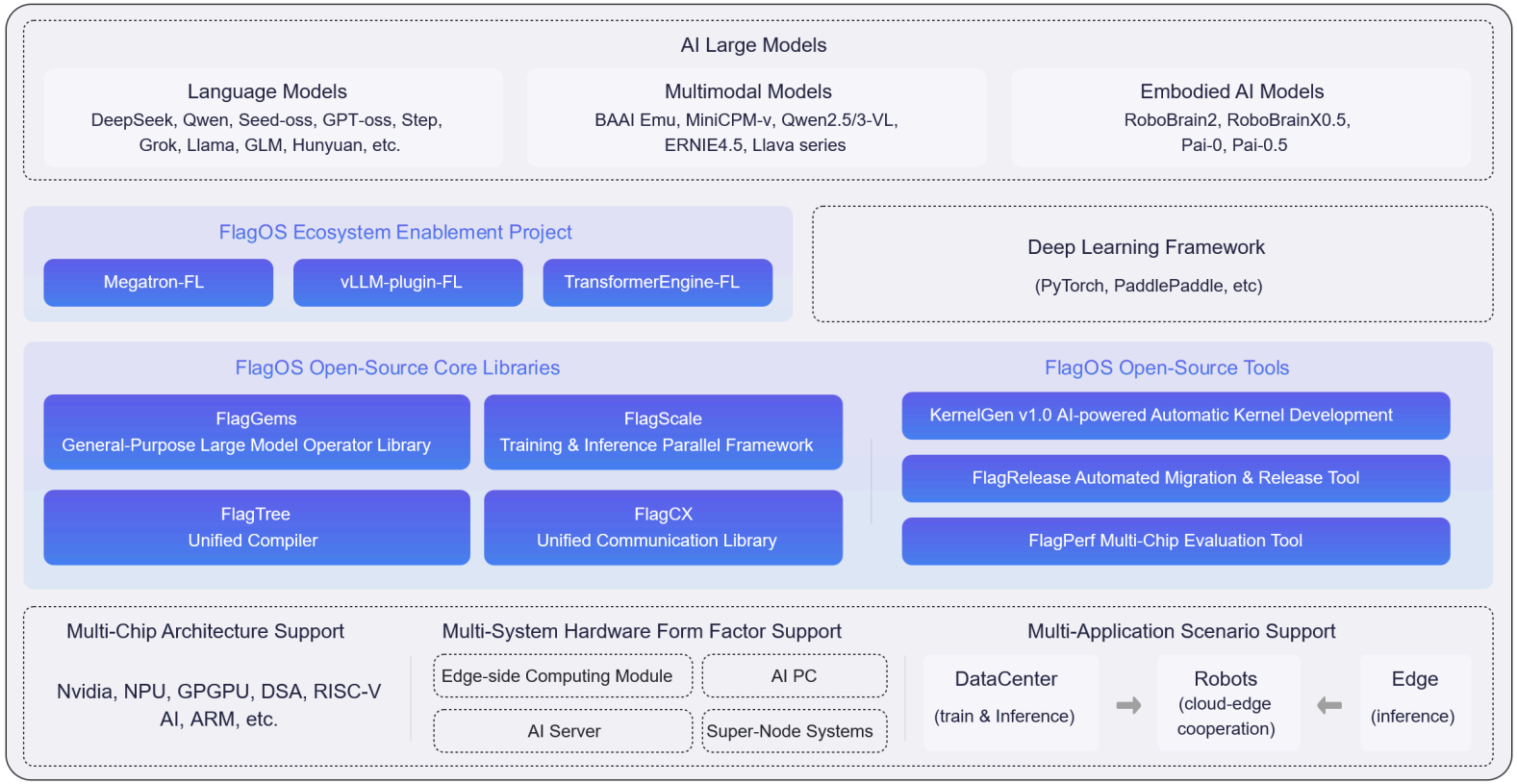
1.5. FICS in CAAFDP

Course: Hardware & Software Fundamentals of Intelligent Computing Systems (FICS)

- Audience: faculty developing AI-compute curricula
- Goal: a teachable **full stack** — device → operators → compiler → distributed → perf / kernelgen
- Practice stack: FlagOS (FlagGems / FlagTree / FlagCX / FlagScale) + non-NV accelerators

Build intuition with numbers (Roofline, memory, collectives), then transfer the same mental model across CUDA / CANN / Triton / FlagOS.

1.6. FlagOS: a Unified AI Software Stack



2. Modules

- Five-module map & October schedule
- Module previews
  - hetero → ops → compiler → dist → perf / kernelgen

2.1. Five Modules at a Glance

Module	Focus
1 · Hetero systems	Device / host parallelism; Triton programming basics
2 · AI operators	Attention, FlashAttention, fusion, operator libraries
3 · AI compilers	IR, `torch.compile`, Triton → TTIR / TTGIR / LLVM
4 · Distributed	DP / TP / PP; collectives & topology
5 · Perf & KernelGen	Metrics, Roofline, autotune, AI-assisted kernels

- same spine as the undergraduate AI system-centric course
- compressed for a faculty development program.

2.2. Program Logistics

AI Compute Faculty Development Initiative — Oct 12–25, 2026

(14 days total; 10 instructional days)

Item	Detail
Hours	AM 09:00–12:00 · PM 14:00–17:00 (UTC+8); 15-min break each half-day
Language	Instruction in English; slides EN + FR
Prep	Bring a laptop; online lab environments are pre-configured
Meals	Lunch on instructional days (Days 1–5, 8–12)
Venue	3F, Building C, Zhongzhiyuan, Zhongguancun Dongsheng Science Park, Haidian

Legend: L Lecture (instructor) · P Practice (TA-guided) · D Discussion / demo

Design: **lab → theory** — build intuition first, then formalize.

2.3. Week 1 Schedule (Oct 12–16)

Day	Date	Theme	AM	PM
1	Mon 12	Opening & overview	D Opening & icebreaking · course overview	D General course on LLMs · P Inference lab (full pipeline)
2	Tue 13	Inference & LLM	D Edge model vendor & edge device	P Lab: Triton primer
3	Wed 14	AI systems & edge	P Edge device labs	L Module 1: hetero systems theory
4	Thu 15	Kernels & talent	P Lab: fused Add + RMSNorm	D AI4Good Global Exchange Forum
5	Fri 16	Operators & visit	L Module 2: high-perf operators	D China tech & culture · P Profiler demo

Sat–Sun 17–18: weekend · D FlagOS Developer Conference

2.4. Week 2 Schedule (Oct 19–23)

Day	Date	Theme	AM	PM
8	Mon 19	KernelGen & perf	P KernelGen lab · L Module 5 theory	D Vendor talk · UNU talk · business exchange†
9	Tue 20	Profiling & compilers	P Company visit 1 (Tencent / Huawei)	L Module 3 theory + P compiler lab
10	Wed 21	Distributed training	P Lab: distributed training (FlagScale)	L Module 4 theory (part 1/2)
11	Thu 22	Dist. & industry	L Module 4 theory (part 2/2)	P Company visit 2 (Xiaomi / Huawei)
12	Fri 23	Pedagogy & close	D Pedagogy discussion	D Closing ceremony

† Oct 19 PM extended 14:00–18:00. · Sat–Sun 24–25: free time.
Schedule may adjust slightly for guests and visits.

2.5. FICS Theory Slots

Instructor-led **L** **Module** theory blocks in the program calendar:

When	Module	Title
Wed Oct 14 PM	01	AI Systems Fundamentals & Heterogeneous Computing
Fri Oct 16 AM	02	High-Performance AI Operators & Operator Engineering
Mon Oct 19 AM	05	Performance Benchmarking & Next-Gen Kernel Generation
Tue Oct 20 PM	03	AI Compiler Principles & Optimization (+ lab)
Wed Oct 21 PM	04	Distributed Parallel Training (part 1/2)
Thu Oct 22 AM	04	Distributed Communication Primitives (part 2/2)

Also join:

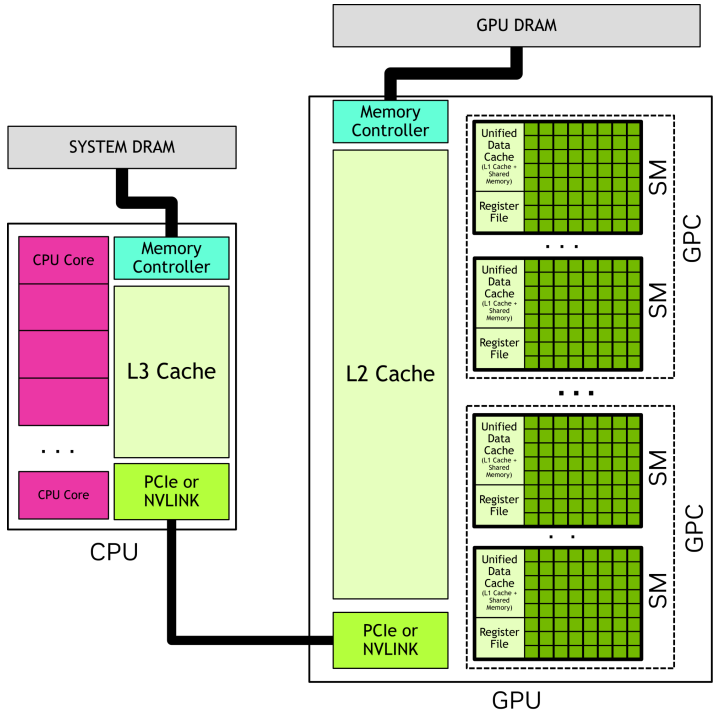
- **Mon Oct 12** — opening & course overview
- **Thu Oct 15 PM** — AI4Good forum
- **Fri Oct 23** — pedagogy & closing

2.6. Module 1: Systems & Heterogeneous Computing

- AI hardware landscape: cloud / edge / mobile
- Host–device model; PCIe / NVLink-class links
- Case study: Iluvatar BI-V150
- Programming bridge: **Triton** as the course lingua franca



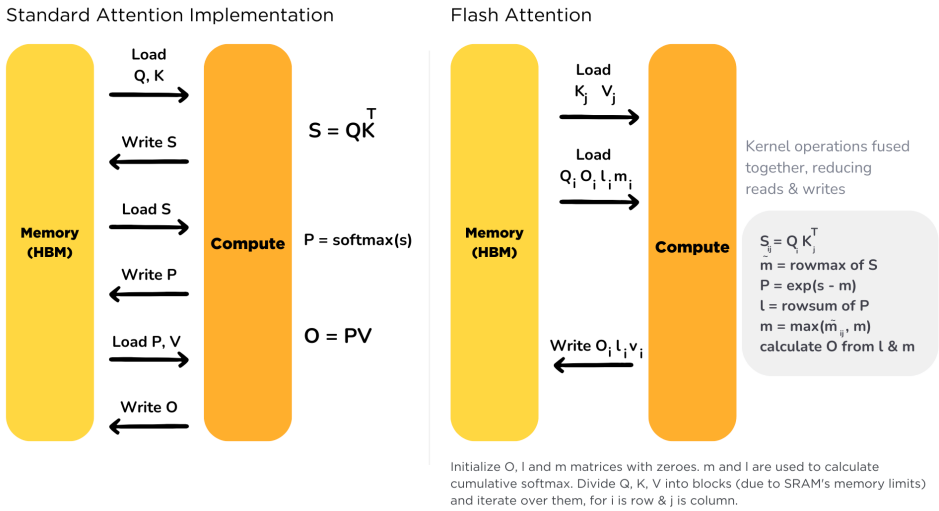
Cloud / edge / mobile AI hardware



Host–device: CPU ↔ GPU via PCIe / NVLink

2.7. Module 2: High-Performance Operators

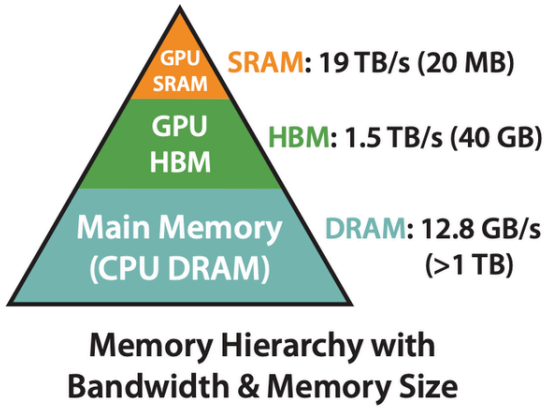
- Transformer operators as a systems workload
- Attention I/O wall → FlashAttention tiling & fusion
- Operator fusion case study (RMSNorm + residual)
- Libraries & dispatch (FlagGems / vendor libs)



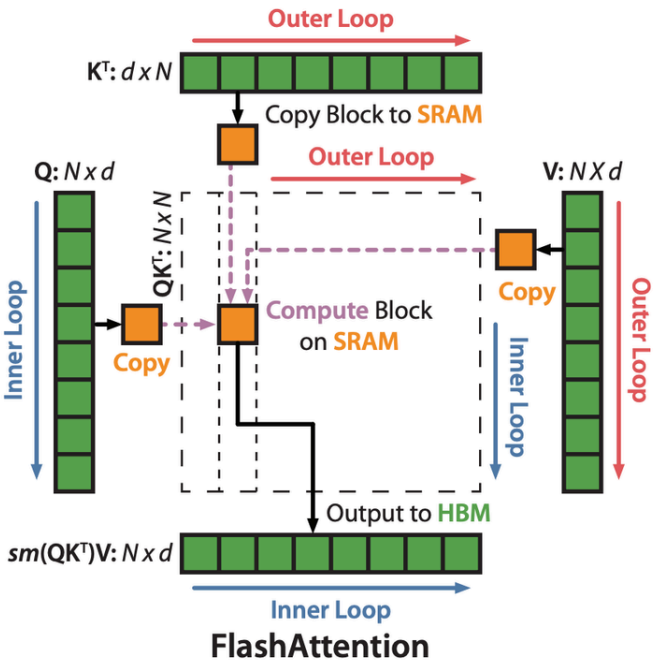
Standard attention vs FlashAttention I/O

2.8. Module 2: Less Data Movement

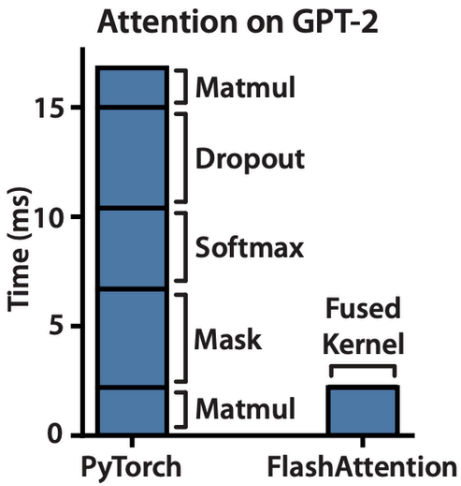
- Standard attention writes huge S, P matrices to HBM
- FlashAttention tiles into on-chip SRAM and fuses softmax + matmul
- Same math, far less data movement → memory-bound → bandwidth-friendly



Memory Hierarchy with Bandwidth & Memory Size



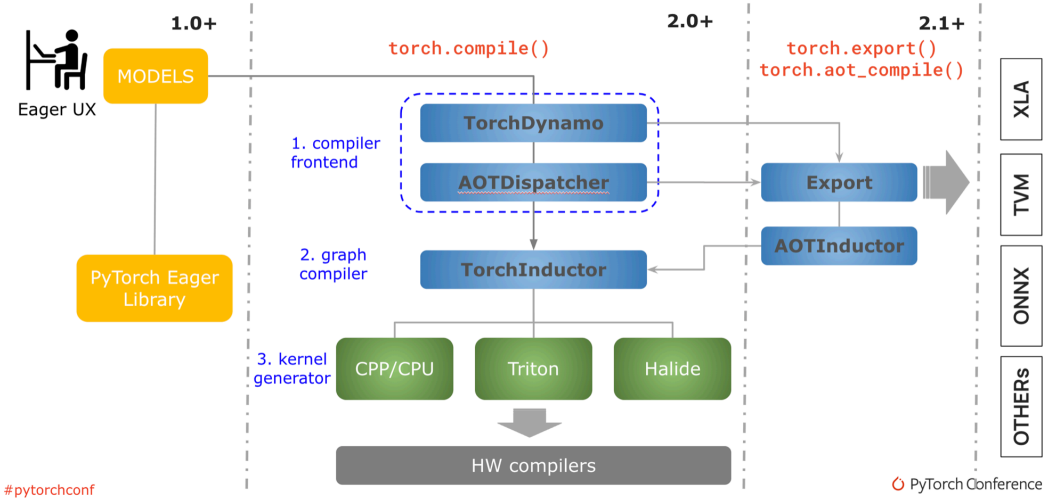
FlashAttention



2.9. Module 3: AI Compilers

- Graph capture: FX / `torch.compile`
- Triton path: eDSL → TTIR → TTGIR → LLVM / backend
- Passes that matter: fusion, layout, memory, codegen

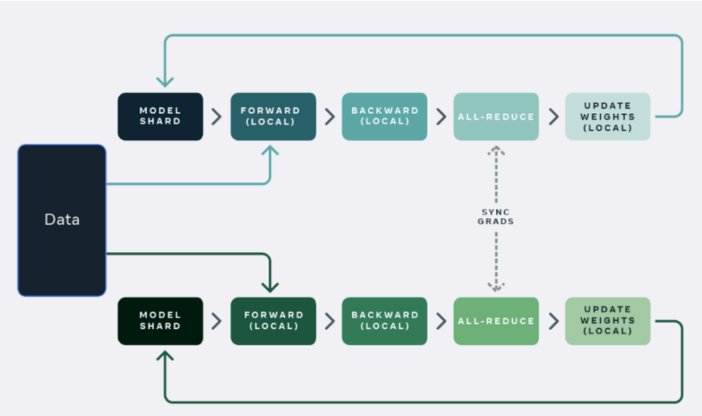
Operators are **what** to compute; compilers decide **how** to map tiles, stages, and memory.



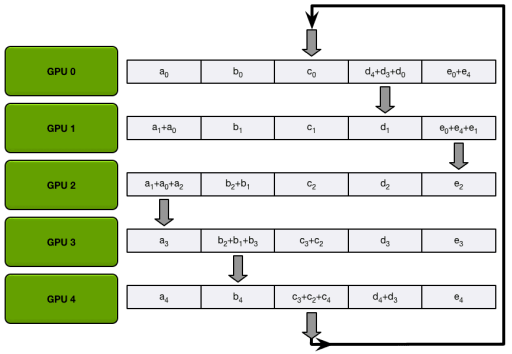
`torch.compile`: Dynamo → Inductor → Triton / backends

2.10. Module 4: Distributed Training & Collectives

- Data Parallel (DP) / ZeRO–FSDP; Tensor Parallel (TP); Pipeline Parallel (PP); hybrids
- Collectives: Broadcast, Reduce, AllReduce, AllGather, ReduceScatter
- Topology-aware algorithms on real clusters



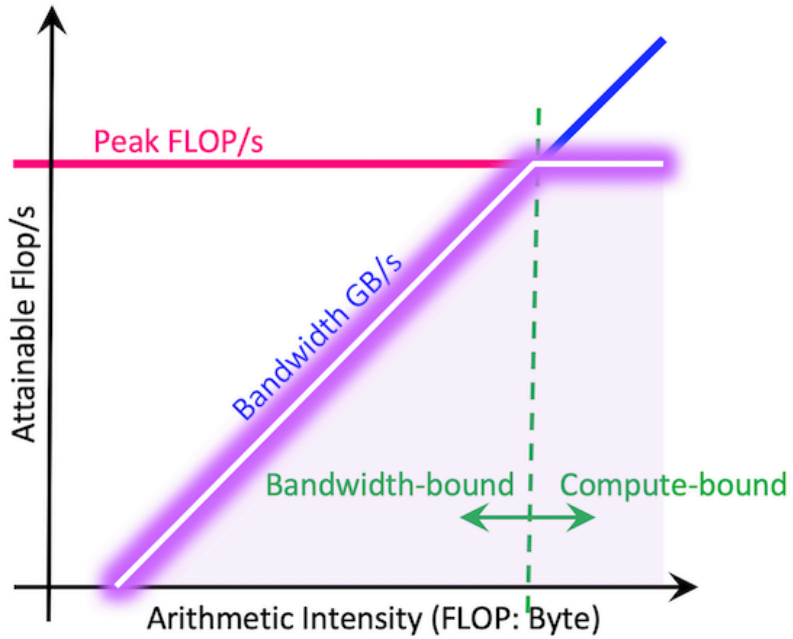
Data parallel: local fwd/bwd + AllReduce grads

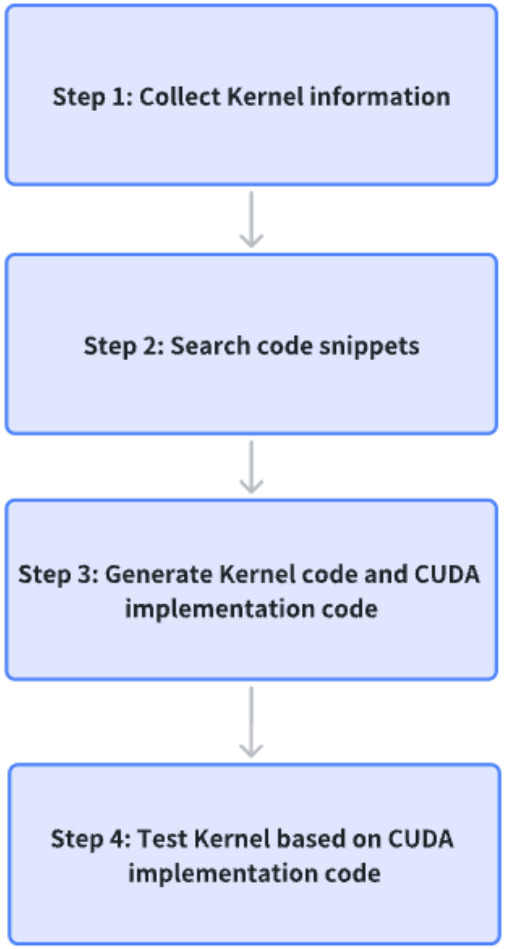


Ring AllReduce — one communication step

2.11. Module 5: Performance & KernelGen

- Metrics: FLOPS, bandwidth, latency, utilization
- Roofline as the shared diagnosis language
- Autotuning search spaces
- KernelGen: generate → test against baseline → report speedup





3. Practice & Close

- Labs & how to reuse the decks
- Takeaways & resources

3.1. Labs & Teaching Reuse

Lab themes

- Operator / GEMM — measure, fuse, tune
- Compiler path — IR dumps; fused vs unfused
- Distributed micro-experiments — DP vs TP cost
- Perf diagnosis — Roofline before “optimizing blindly”

Deck reuse

- Module decks (`mod01`–`mod05`): half-day faculty sessions
- Unit decks (`l01`–`l48`): semester-scale classrooms
- EN + FR twins where needed

Prefer **one measured question per lab** over a long recipe dump.

3.2. Takeaways

1. **Numbers matter** — effective compute and bandwidth must advance together.
2. **Organize by system layers** — hetero → ops → compile → distribute → measure/generate.
3. **One programming bridge** — Triton / tile kernels across vendors.
4. **Practice in FlagOS** — open tools faculty can reassign and extend.
5. **Diagnosis before knobs** — Roofline and collectives explain “why” before “how”.

3.3. Resources & Closing

- Program schedule: AICFDI 2026h2 schedule
- Labs: AICFDI 2026h2 labs
- FlagOS: github.com/flagos-ai
- KernelGen: github.com/flagos-ai/KernelGen

Treat the AI cluster as a computer —
and give students a stack they can program, measure, and extend.

African Scientific Research and Innovation Council

BAAI 智源研究院

北京大学 PEKING UNIVERSITY

中国人民大学 RENMIN UNIVERSITY OF CHINA

Thanks

Questions & further discussion welcome

Guojie Luo
罗国杰
gluo@pku.edu.cn

Jing Wang
王晶
jwang@ruc.edu.cn

BAAI Team
智源团队
github.com/flagos-ai



Supported by UDNP.